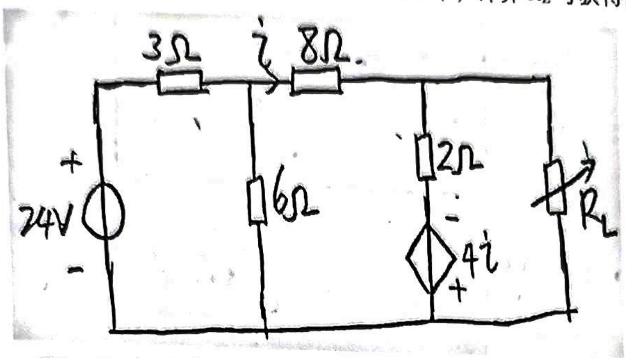
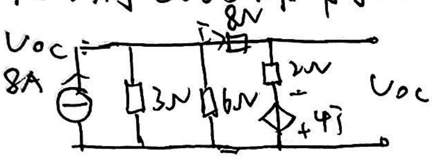
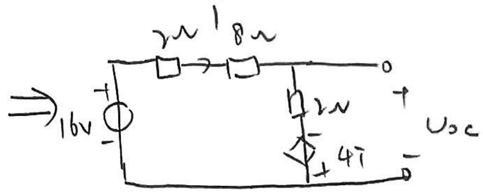
1.如图 3-1 所示电路，(1) R_L 为何值时，可获得最大功率；(2) 计算 R_L 可获得的最大功率。



对R_L左侧电路戴维南等效：

求U_{oc}：





$16-10i=-4i+2i$ $i=2A$ $U_{oc}=-4V$

求I_{sc}：

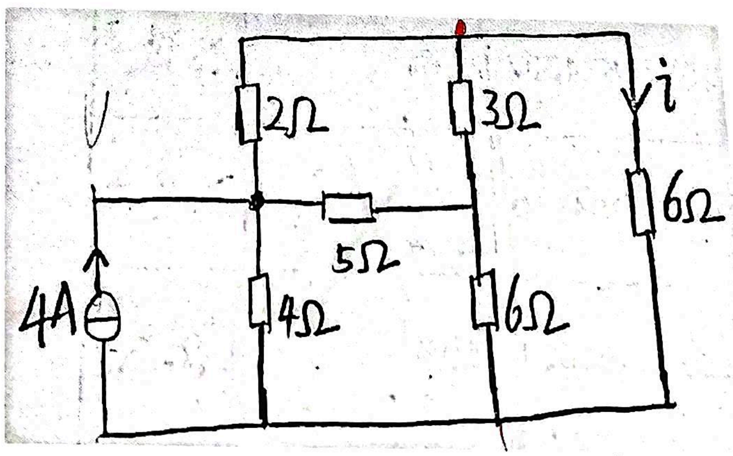
$I_{sc}=1.6A$

$R_i=\frac{U_{oc}}{I_{sc}}=2.5Ω$

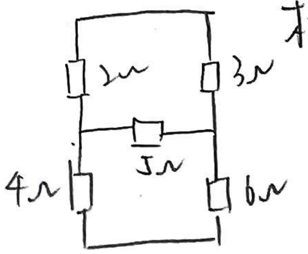
$R_L=R_i=2.5Ω$时

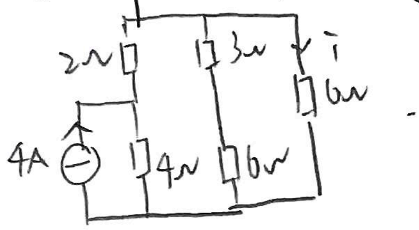
$P_{max}=\frac{U_{oc}^2}{4\cdot R_L}=1.6W$

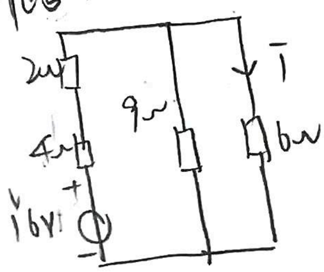
2.如图 3-2 所示电路，计算电流 i。



构成电桥：5Ω电阻相当于断路：

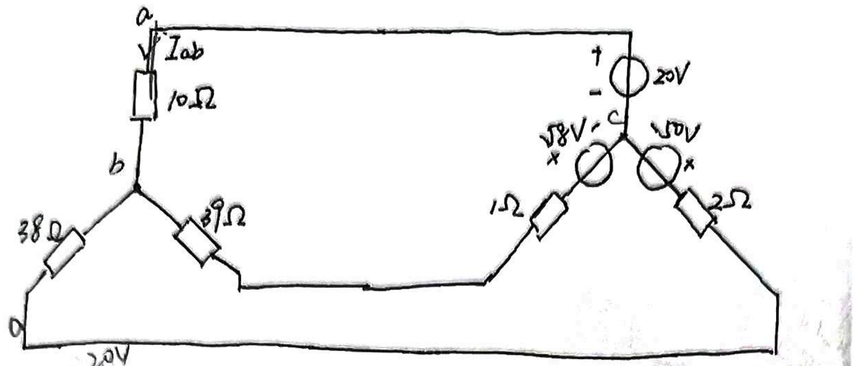


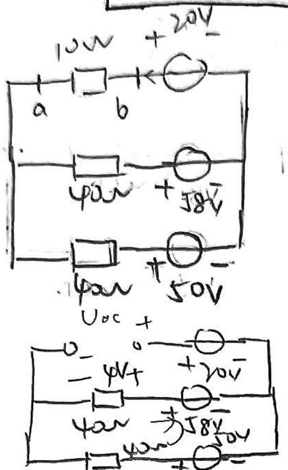




$$i=\frac{16}{9/\!/6+6}\times\frac{9}{9\times 6}=1A$$

3. 试用戴维南定理求题图 3-3 所示电路中 ab 支路的电流 I_{ab}





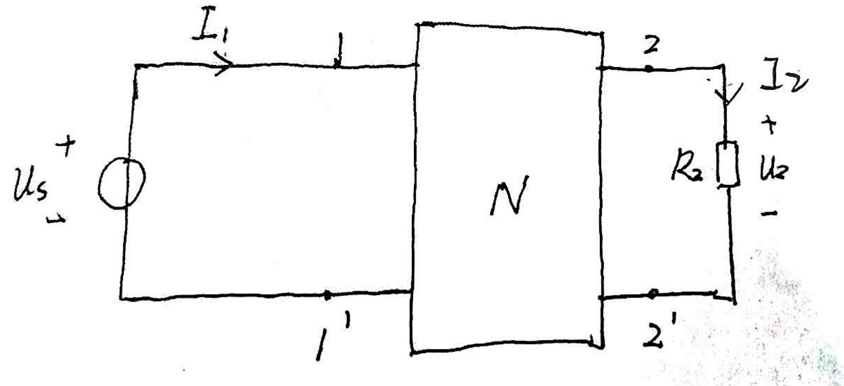
对ab外支路戴维南等效

$R_i = 40 \parallel 40 = 20\Omega$

$U_{oc} = -58 + 4 + 20 = -34V$

$\therefore I_{ab} = \frac{-34}{20+10} = -1.13A$

4. 题图 3-4 所示电路中，N 为无源线性电阻网络。当 $R_2 = 2\Omega$，$U_s = 6V$ 时，测得 $I_1 = 2A$，$U_2 = 2V$。如果当 $R_2 = 4\Omega$，$U_s = 10V$ 时，又测得 $I_1 = 3A$，求此时的 U_2。



$R_2 = 2\Omega$ 时 $U_S = 6V$ $I_1 = 2A$ $U_2 = 2V$ $I_2 = \frac{2}{2} = 1A$

$R_2 = 4\Omega$ 时 $U_S' = 10V$ $I_1' = 3A$ $U_2' = ?$ $I_2' = \frac{U_2'}{4}$

$U_S'(-I_1) + U_2' I_2 = U_S \cdot (-I_1') + U_2 (I_2')$

$6 \times (-3) + 2 \times \frac{U_2'}{4} = 10 \times (-2) + U_2' \times 1$

$U_2' = 4V$